

Bioinformatic approaches to regulatory genomics and epigenomics

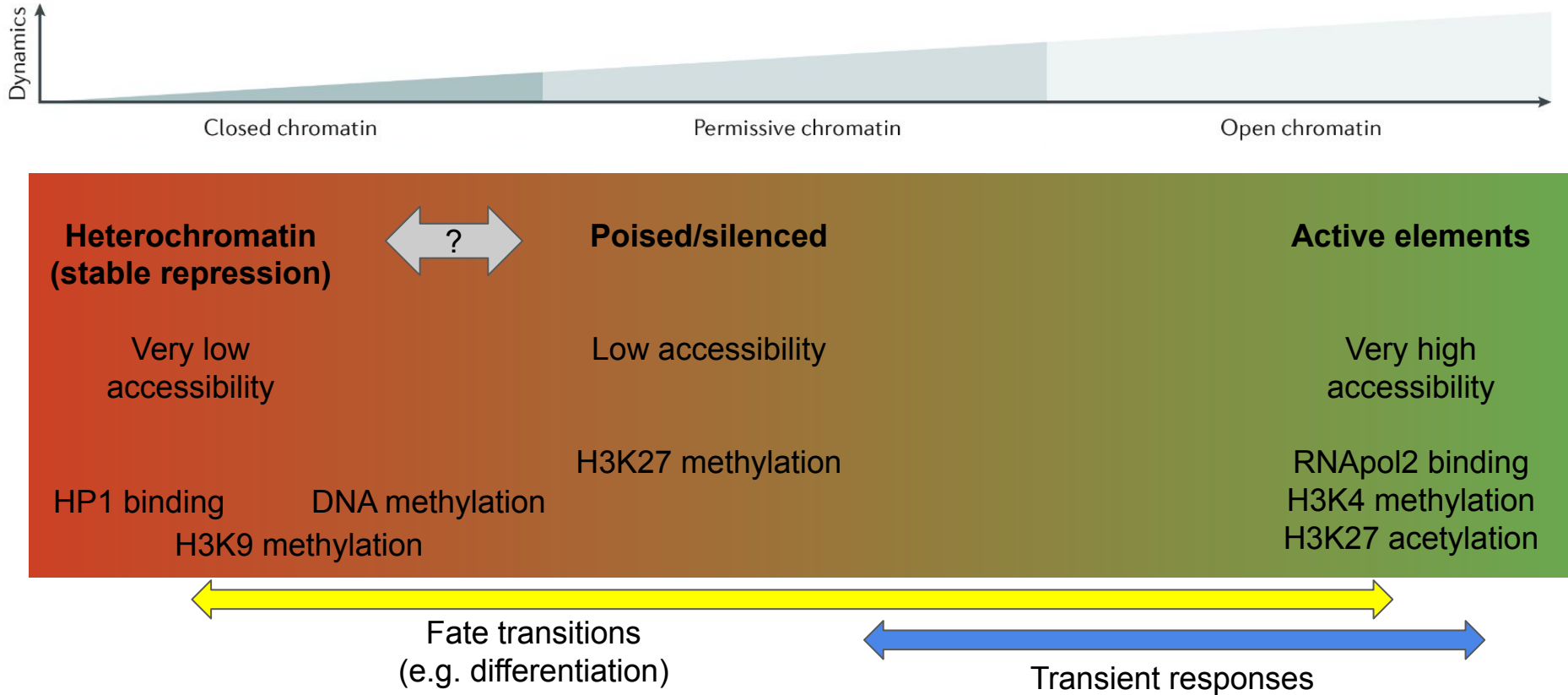
376-1347-00L | week 9

Pierre-Luc Germain

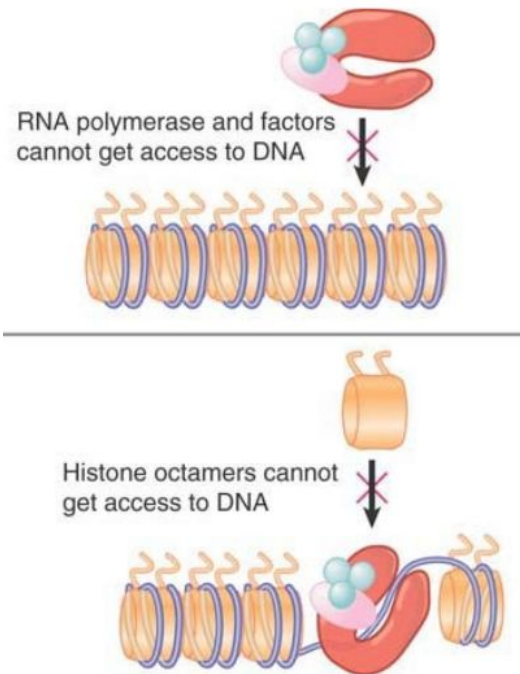
Plan for today

- Theory: from repression to activation and back
- Clustering on genomic signals
- Region-based GO enrichment analysis

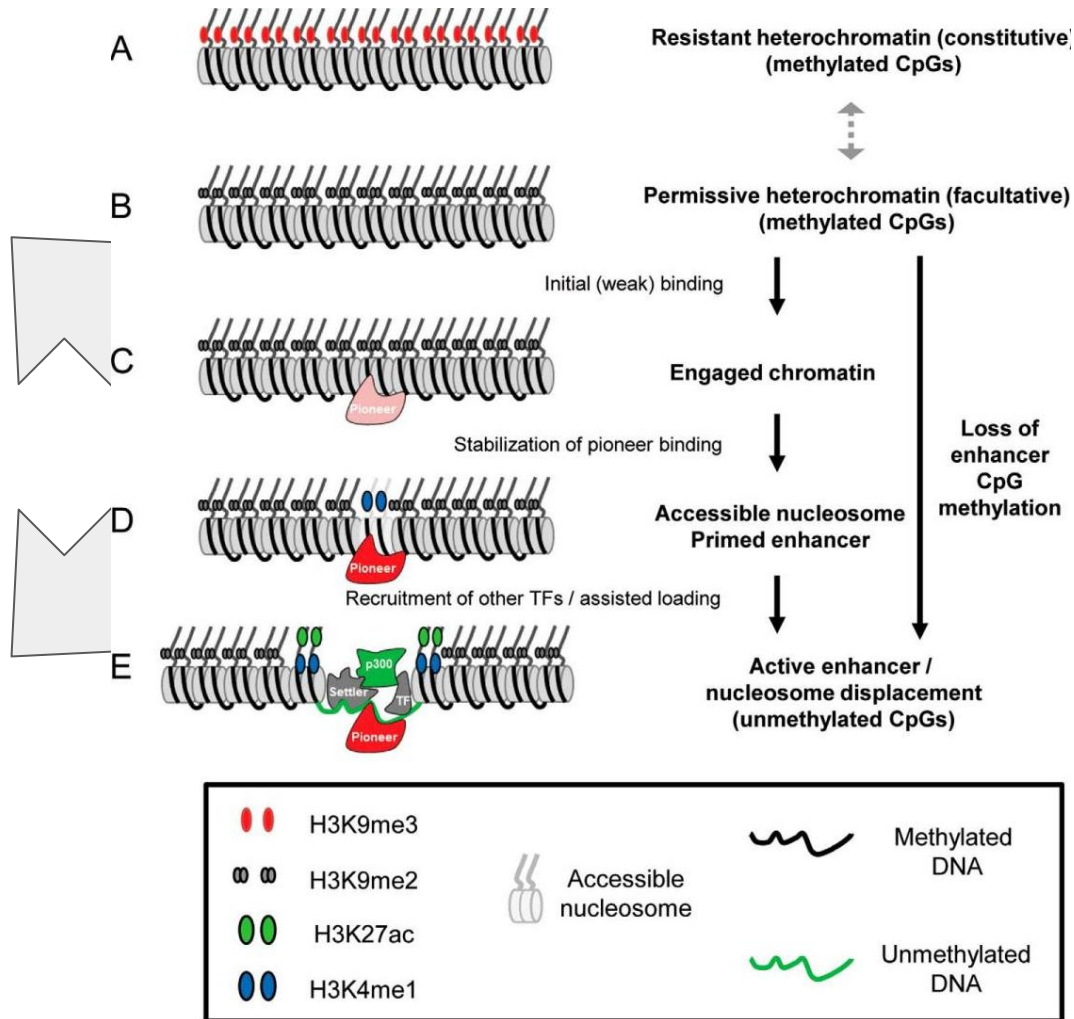
There are degrees of accessibility (activation/repression)



Opening chromatin



(Adapted from Krebs, Goldstein and Kilpatrick, Genes XII, 2018)



(Mayran and Drouin, J Biol Chem 2018)

Whereas most TFs cannot bind their target DNA when it's wrapped around nucleosomes, **pioneer factors** can, and can even bind in heterochromatin

Their binding is typically (but not always) independent of (and prior to) other factors

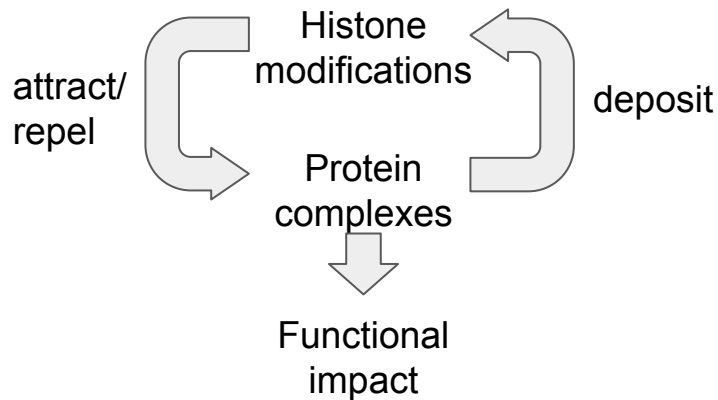
(Adapted from
Mayran and Drouin, J Biol Chem 2018)

Factor	Binding to heterochromatin	Chromatin activation	Epigenetic memory: DNA demethylation	Cell fate reprogramming	Nucleosome binding
Ascl1/Mash1	102	102		102, 103	
C/EBPα		43		104	
Ebfl	47, 48	47, 48	48		
Esrrb					
Foxa	3	3, 4, 28, 31, 32	4, 69, 85	38, 39	28
Gata	59	59		38, 39	29
GR/AR	18	18			
Klf4	21, 22	21, 22		106, 107	53
Neurod1				50, 70	
Nrf1	70	70	Inhibitory (70)		
Oct4	21, 22	21, 22		106, 107	53
p53	100, 101	100, 101			
Pax7	44, 45	44, 45, 61	45	44	
PU.I	41, 42	41, 42		104	
Sox2	21, 22	21, 22		106, 107	53

Opening chromatin: pioneering factors or epigenetic mark?

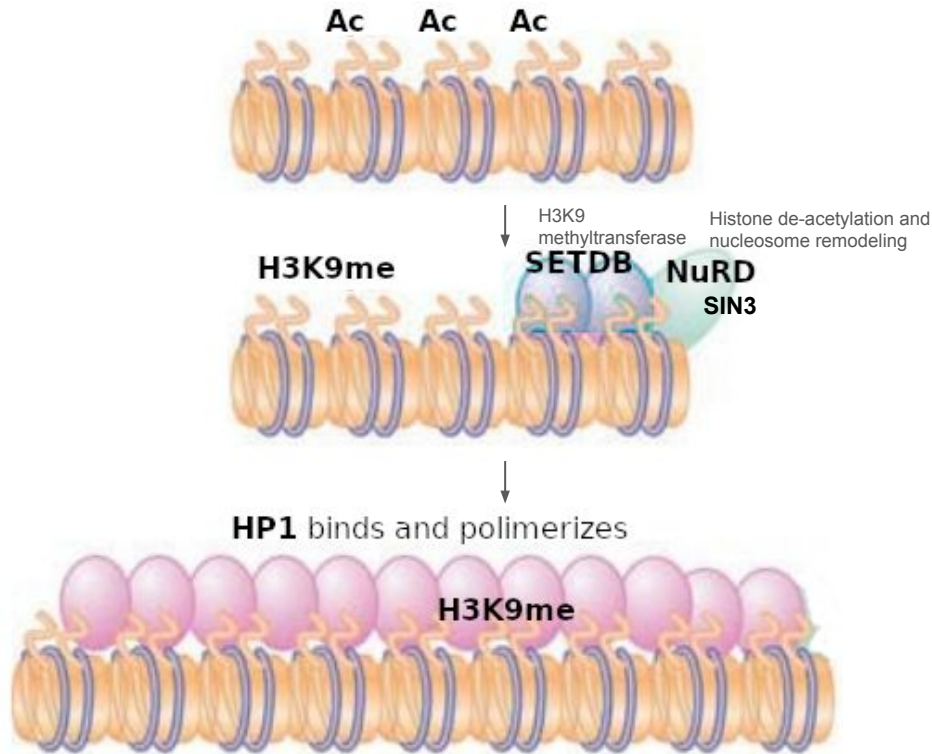
- blocking H3K27ac does not prevent pioneering factors (PF) from opening chromatin
- blocking the PF typically prevents H3K27ac to most sites
- PF binding doesn't always lead to opening, and sometimes require other factors

(Miao et al., Molecular Cell 2022)



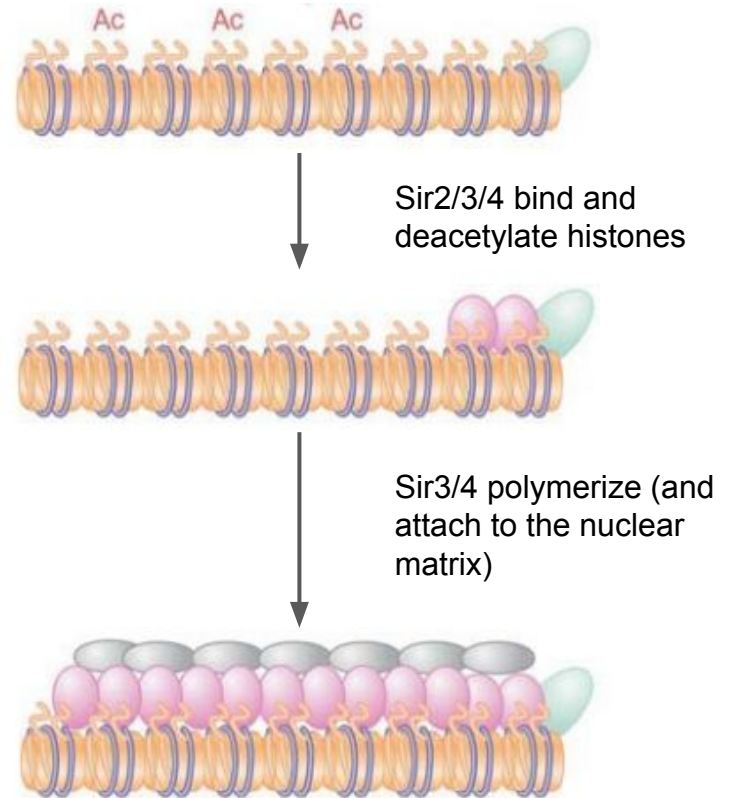
Heterochromatin formation

In mammals:

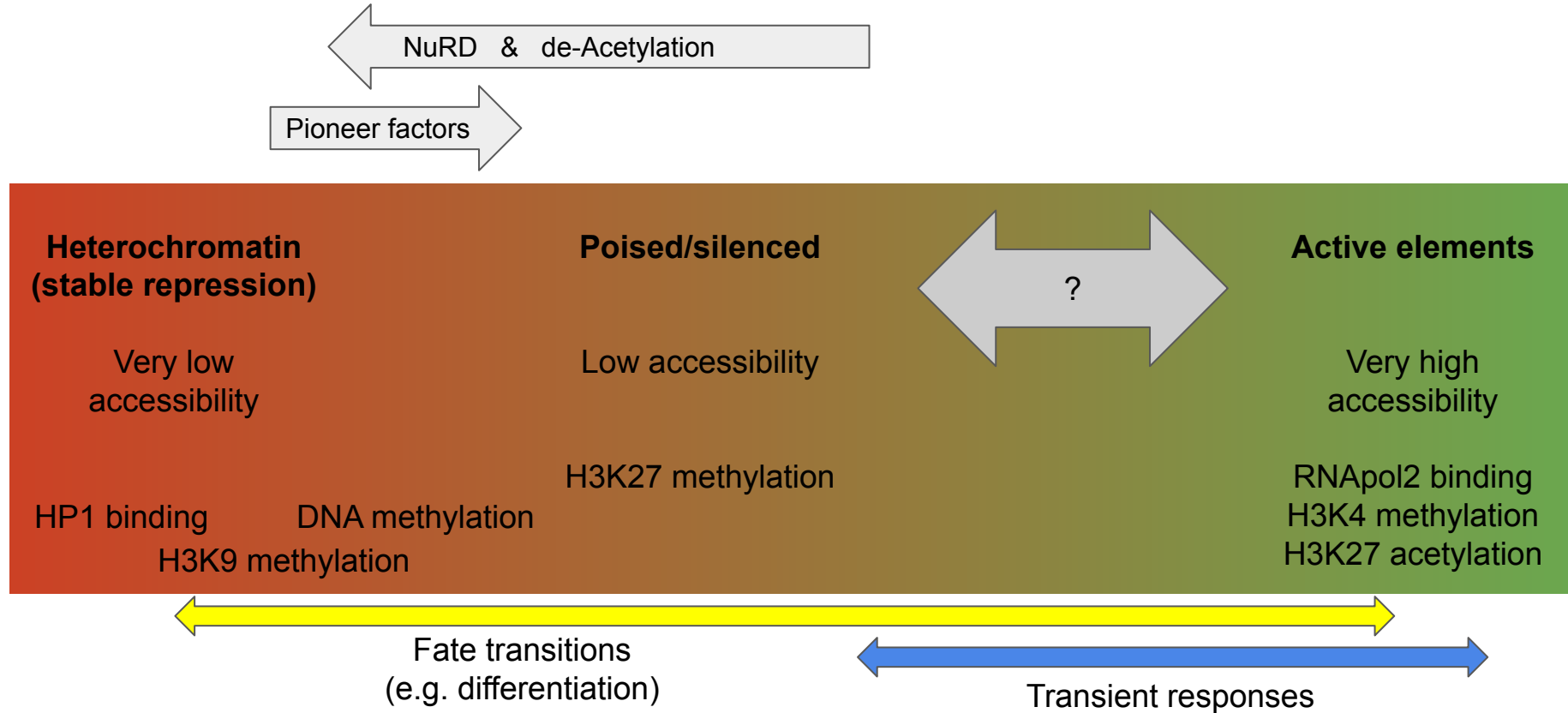


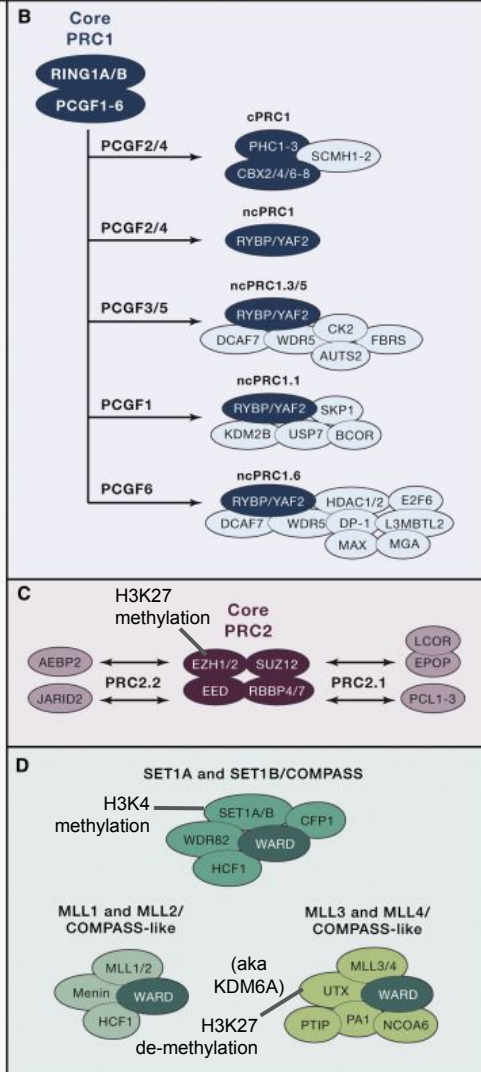
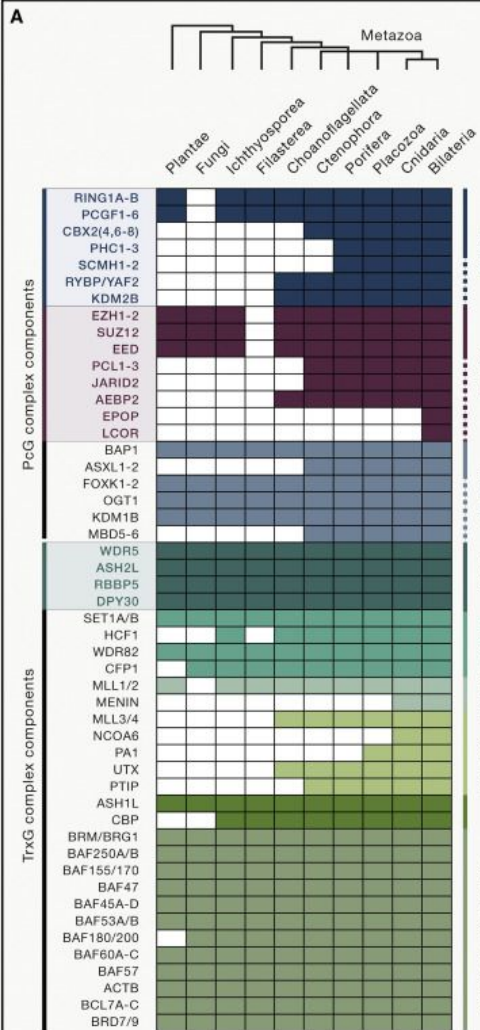
In yeast:

Rap1 binds to DNA

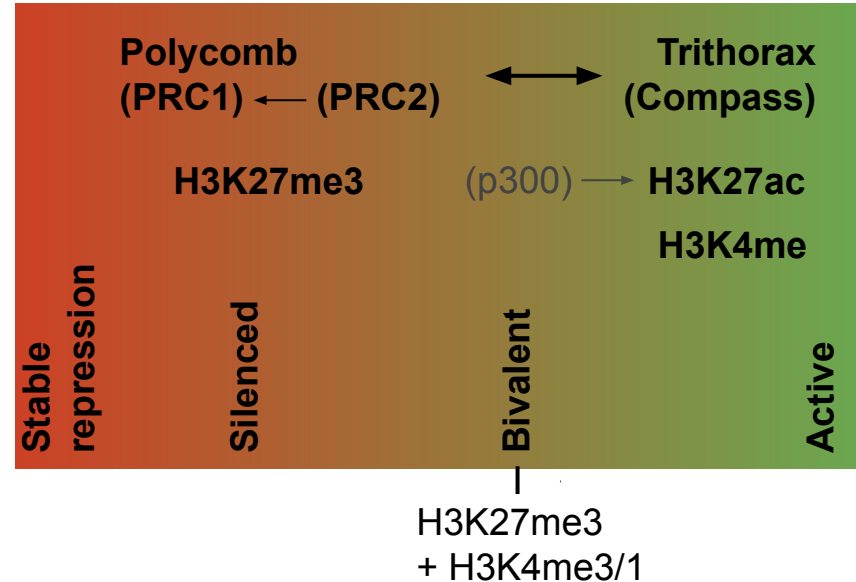


Opening and closing chromatin



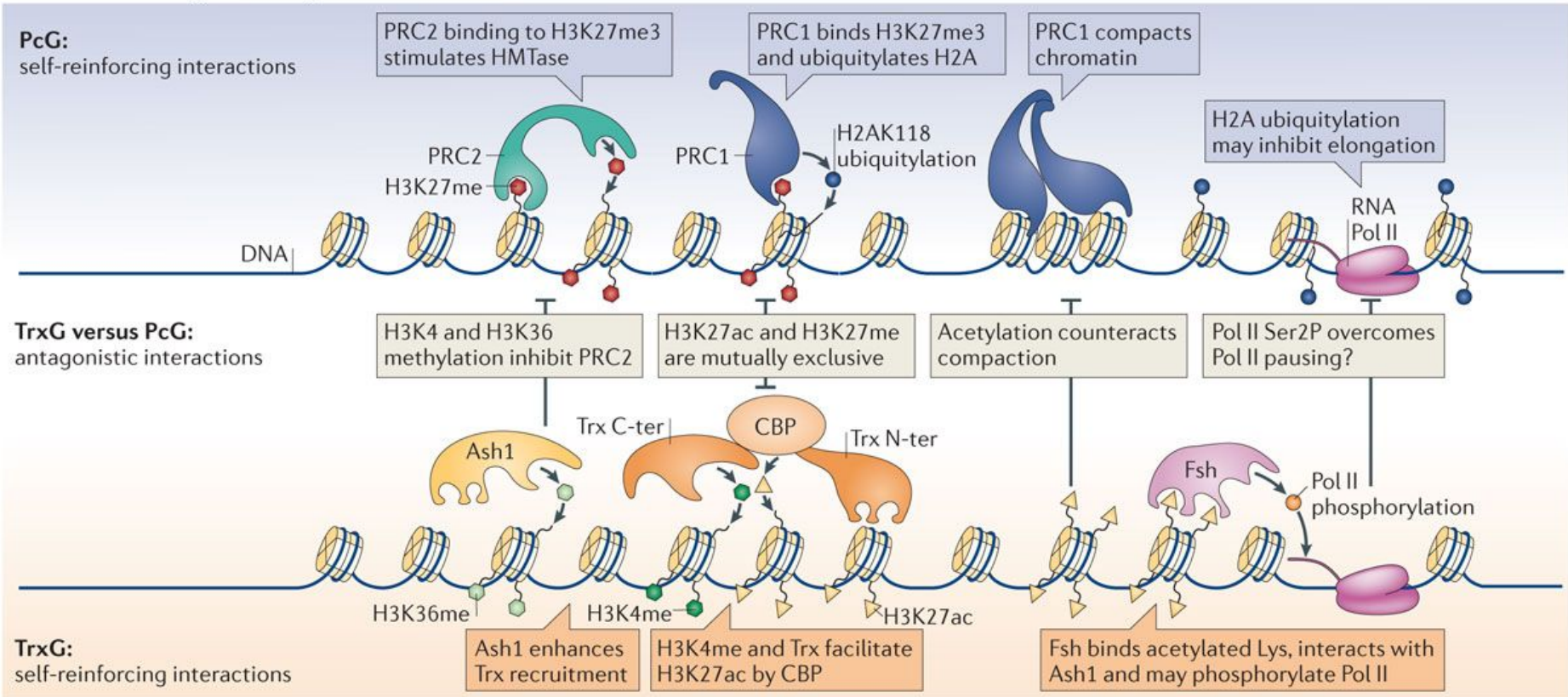


The conserved competition between Polycomb (repressive) and Trithorax (activating) protein groups regulates a very large variety of phenomena

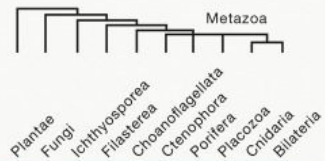


Competition between Polycomb (PcG) and Trithorax (Trx) protein groups

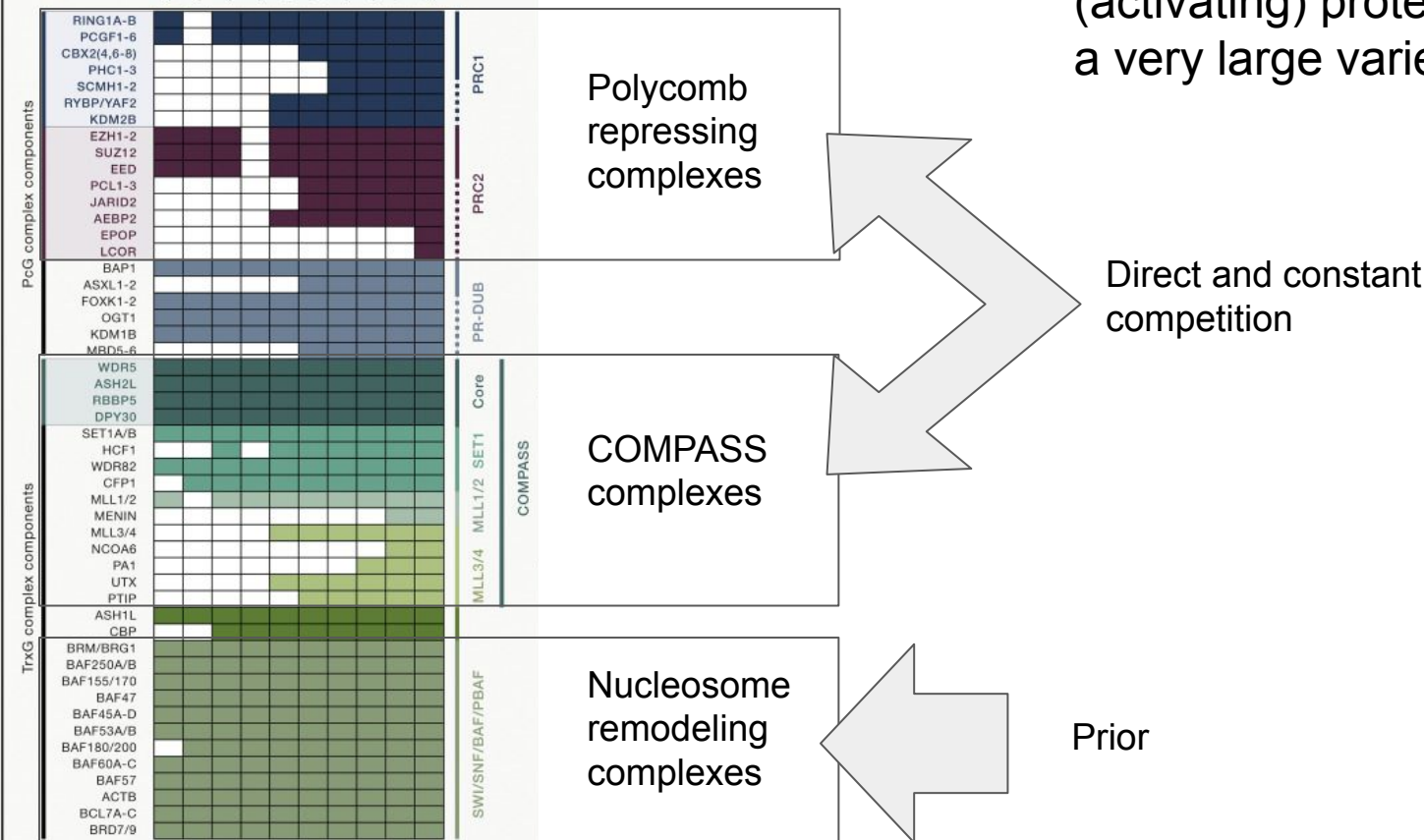
b Self-reinforcing and antagonistic interactions



A

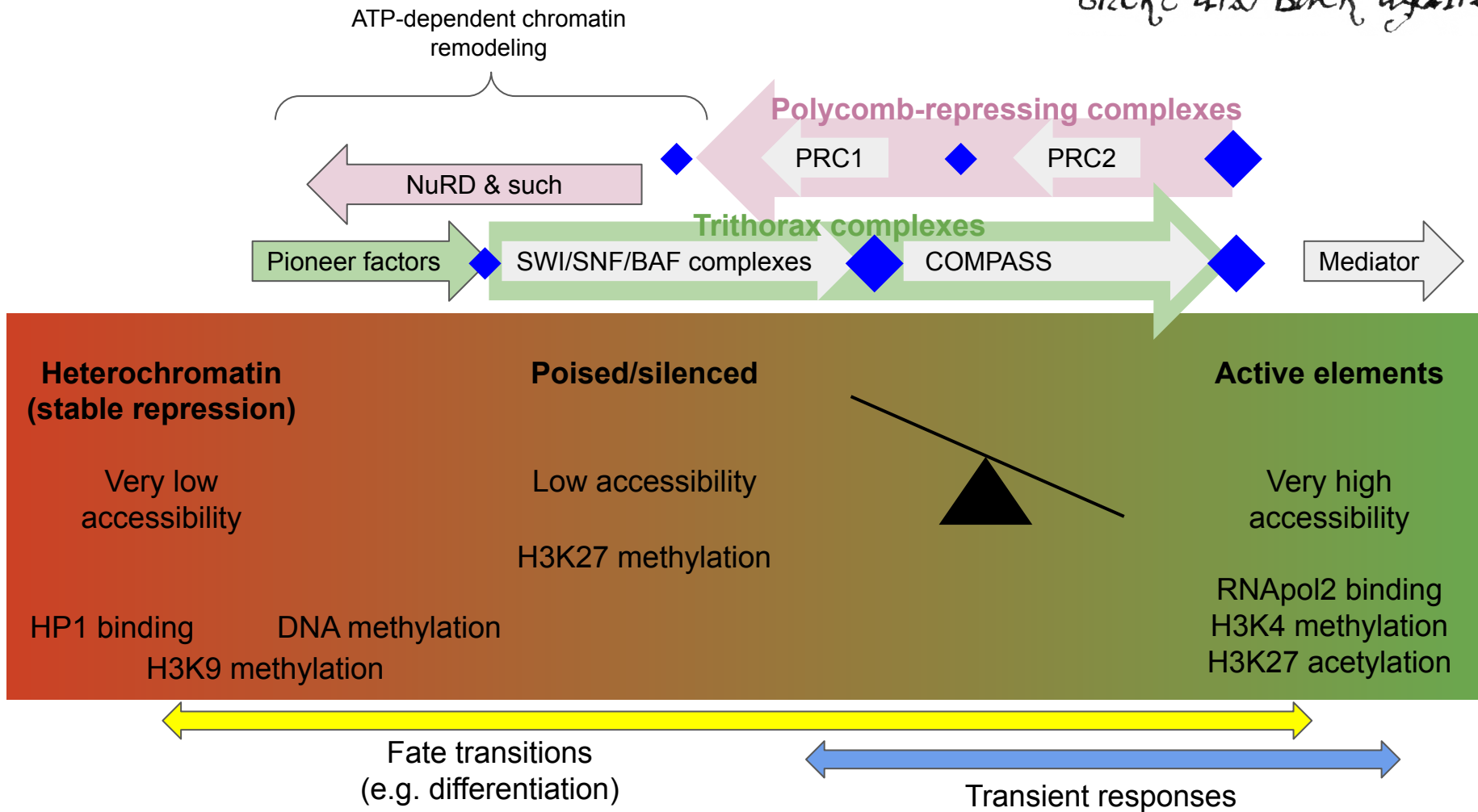


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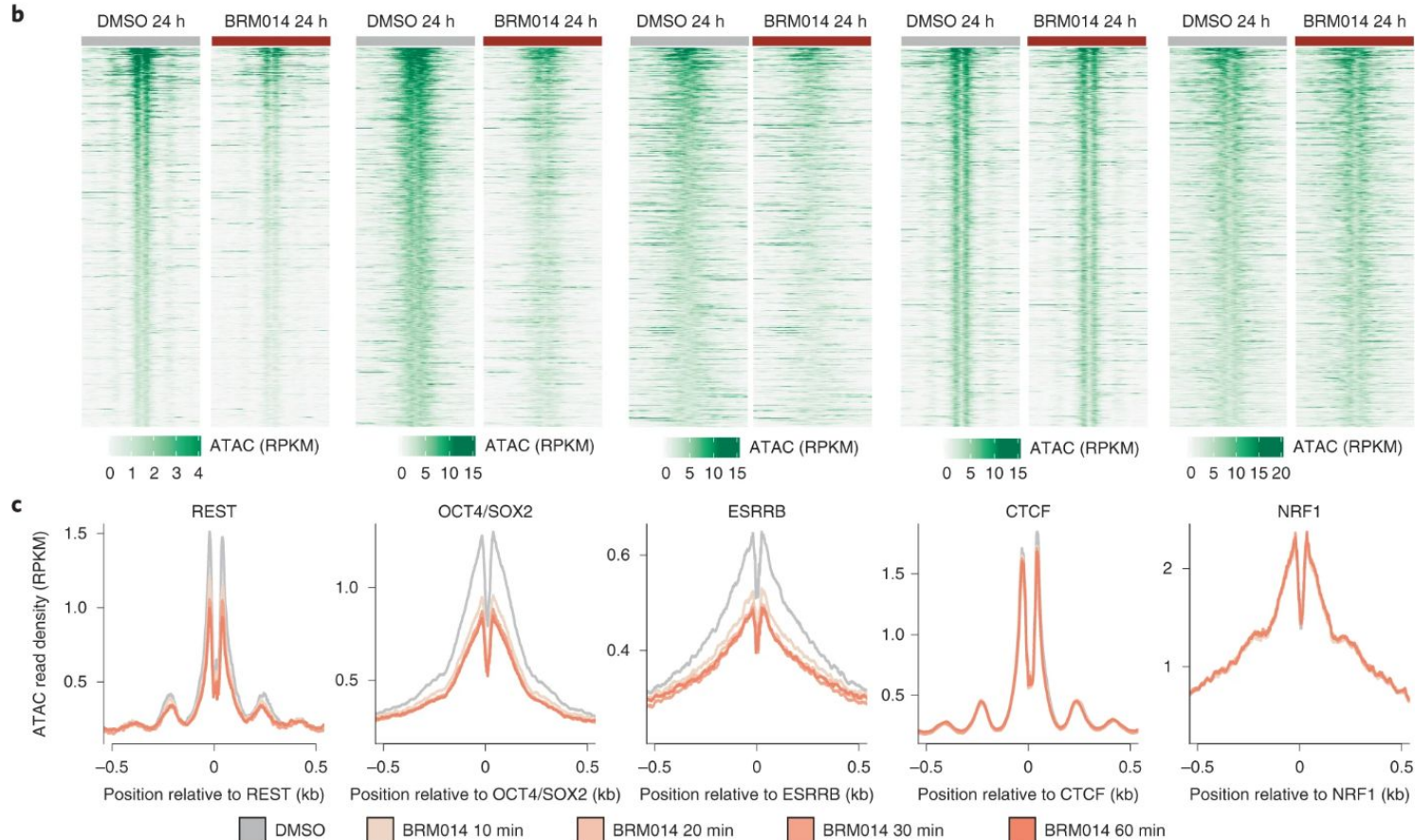


(Schuettengruber et al., 2017)

there and back again...



Inhibition of SWI/SNF activity instantly impairs accessibility at dependent TF-binding sites

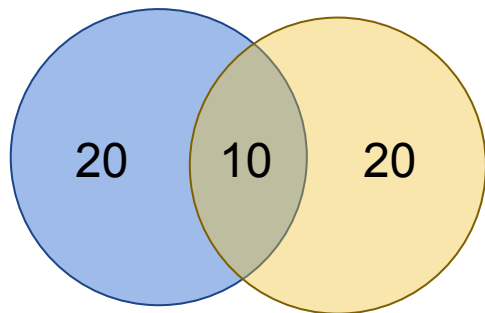


(Adapted from
Iurlaro et al.,
Nat Gen 2021)

Practical:

Clustering epigenomic signals
& GO enrichments in genomic regions

The choice of the background/universe is of critical importance in over-representation analysis!

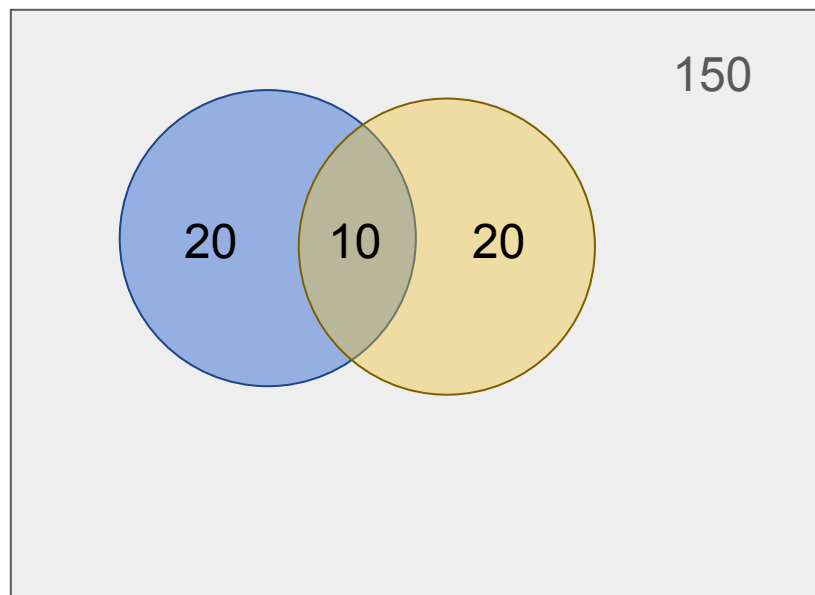


Is this a large overlap between the two sets?

What overlap would you expect by chance?

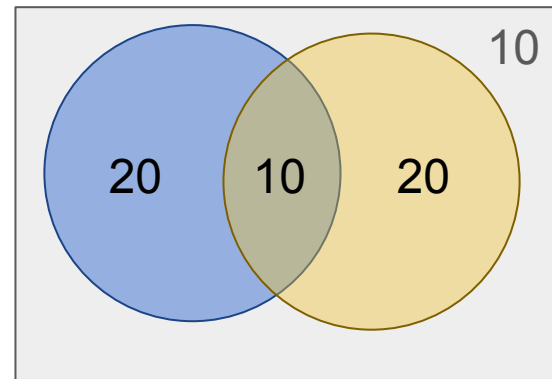
The choice of the background/universe is of critical importance in over-representation analysis!

Universe of 200 elements



Overlap is twice as large as expected ($p=0.01$)

Universe of 60 elements



Overlap is actually much lower than expected!

The choice of the background/universe is of critical importance in over-representation analysis!

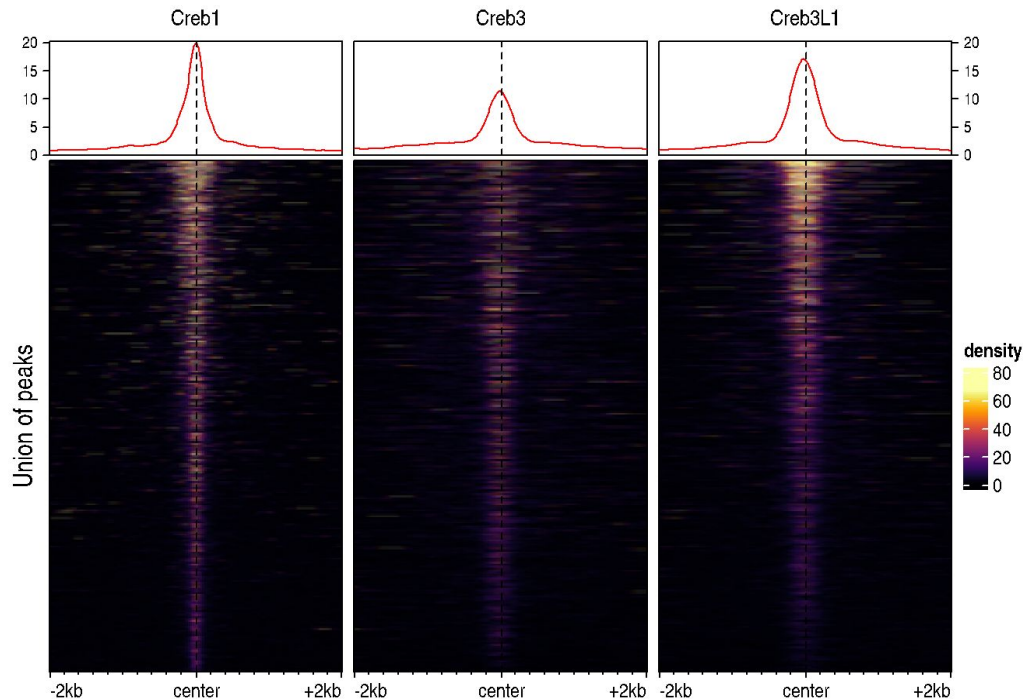
- There is usually not a “good” and “bad” background – what’s important is how the result is interpreted
- Always state what background you used, e.g.
 - Term X is enriched in this set compared to [background]

Assignment

- Download and decompress the following archive:
 - <https://ethz-ins.org/content/w10.assignment.zip>
- This contains the bigwig files and peaks (bed) files for three TFs of the [CREB family](#) (all restricted to chr1; aligned against the hg38 genome)
- Use clustering and visualization to illustrate the relationship between the binding of the different proteins
- Use enrichment analysis (either GO or motif) on at least one of the clusters
- Write a paragraph describing your results
- Save your assignment in a R markdown named `assignment.Rmd`, render it, and push the html file to this folder in your github repository

Assignment

Simply plotting the regions, the signals seem very similar... but are they?



Tip: focus on high-confidence peaks to define the universe of regions, e.g.:

```
peaks <- list.files(pattern="bed$")  
# we first import the peaks  
peaks <- lapply(peaks, rtracklayer::import.bed)  
# we'll focus on the high-quality peaks  
peaks <- lapply(peaks, FUN=function(x) x[x$score>800])  
# we get the union of non-redundant regions  
regions <- reduce(unlist(GRangesList(peaks)))
```